ACTION FAMILY TECH LTD.

FarmCare

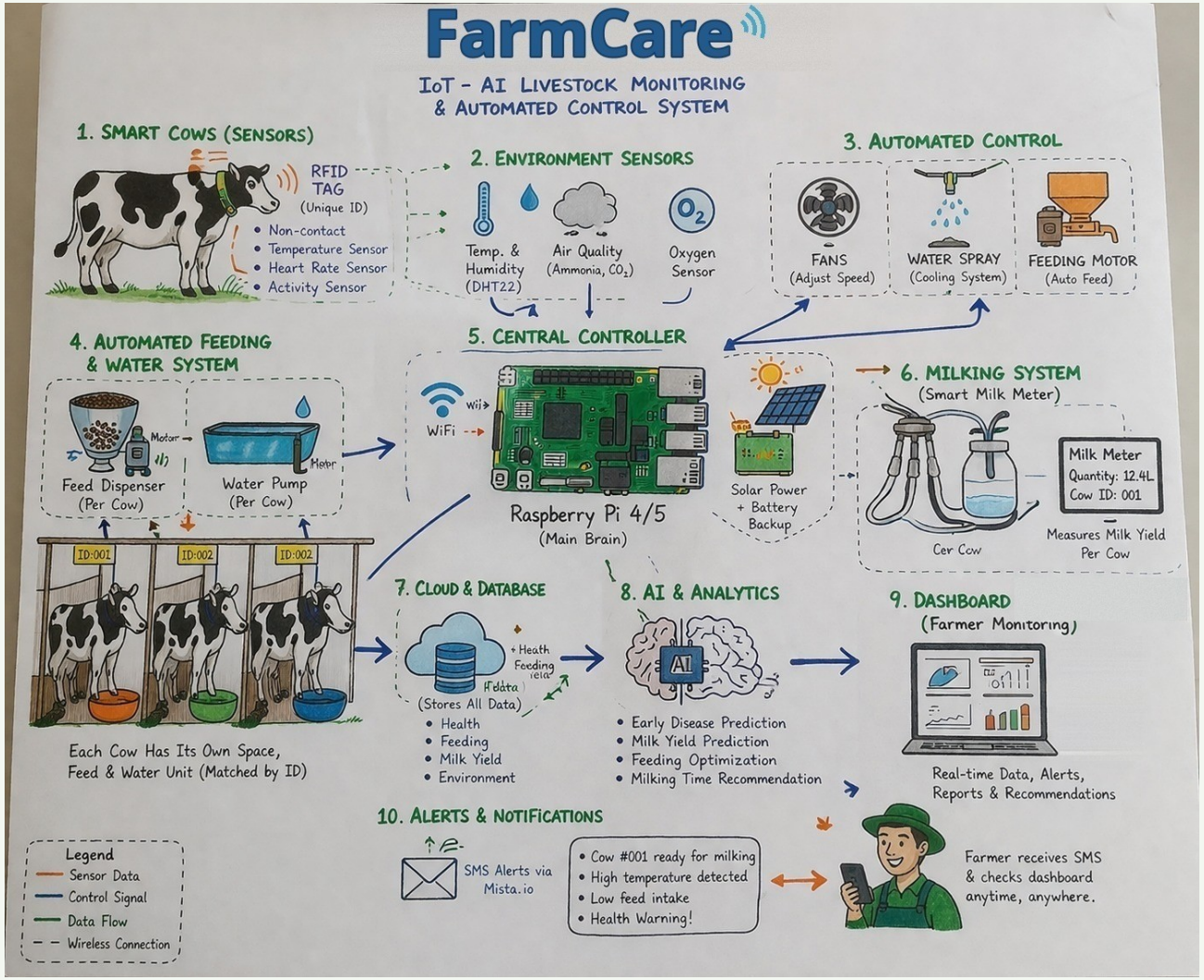
AI and IoT for climate-smart dairy farming

A Milking Station, a Barn Monitor and a Health Collar feed one cloud dashboard, turning everyday farm activity into real-time decisions that protect cows, milk and the climate.

Innovate Africa Challenge III · AI for climate-smart agriculture

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https://farmcareservices.com/



FarmCare system architecture: sensors, control, cloud, AI and alerts

Sick cows, rejected milk, missed pregnancies

1.45–1.73M
cattle across Rwanda's dairy belt

60–67%
of dairy cows have clinical or subclinical mastitis

60%
of milk rejected at collection centres in parts of the North and West

33–50%
insemination success: over half of cows miss pregnancy

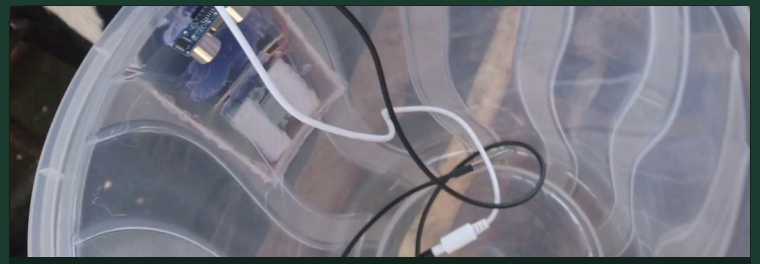
Who is affected: zero-grazing farmers with 1–10 cows in the Eastern Province (Nyagatare, Gatsibo) and the Northern Province (Gicumbi, Burera, Musanze), and the cooperatives and collection centres that absorb rejected milk.

Alternative	What it offers	What it misses
Paper records, vet call-outs	Low cost; the status quo	No sensing: problems found after symptoms; over 95% keep no digital records
Agrotag	Record keeping and traceability tags	No reproduction, feed or live sensing
Difarm	Single-purpose monitoring tools	No link between farm data and collection-centre quality
Imported precision systems	Full sensing and automation	Priced and built for large commercial farms
FarmCare	**Breeding, health, feed, milk quality and barn climate in one platform**	**Links farm data to collection-centre quality and farmer income**

Sources: MINAGRI Annual Report 2024/25; RAB and MINAGRI dairy sector data; competitors' public product information; our own farmer surveys through field visits. Several team members are dairy farmers.

One platform. Three connected devices.

FarmCare pairs wearable and barn sensors with a cloud dashboard and AI analytics, turning everyday dairy-farm activity into real-time decisions.



Milking Station

RFID identifies the cow; an ultrasonic sensor in the milk container measures litres; TDS and temperature probes check quality.

RUNNING IN THE COWSHED



Barn Monitor

Reads heat, humidity, two gas sensors, water-tank level and feed weight; runs pump, fan, water spray and buzzer.

RUNNING IN THE COWSHED



Health Collar

Worn by the cow: body temperature, heart rate, SpO2, motion and live GPS location.

RUNNING IN THE COWSHED

Web platform live

Health, milking and barn dashboards at farmcareservices.com/app

Per-cow risk scoring

Each cow scored against her own 14-day baseline

Field-tested

All three devices in a cowshed in Kanyinya, Nyarugenge

Stage: TRL 6

Working application; demo video shows every module

Climate-smart dairy: adapt, cut emissions, produce more

14.5%

of global greenhouse-gas emissions come from livestock (FAO).

Rwanda's Nationally Determined Contribution under the Paris Agreement targets livestock emissions, while hotter barns cut milk yields.

CH_4 · N_2O · CO_2 · NH_3

Adapt · heat stress

Barn Monitor reads heat and humidity every 10 seconds and switches fans and water spray by itself; the demo shows the spray turning on in the cowshed.

WORKING IN THE COWSHED

Mitigate · manure gases

MQ-135 and MQ-5 sensors read ammonia, CO_2 and methane-type gases from manure and urine live in the cowshed (4–7% of range), flagging build-up early.

LIVE · UNCALIBRATED

Mitigate · biogas

Next: feed captured manure to biogas digesters, with pressure and temperature sensors to optimise methane capture into clean energy.

PLANNED

Produce more per cow

Earlier heat and disease detection means more milk per cow rather than more cows, which lowers emissions per litre.

NOT YET MEASURED

Planned AI: early warning of emission spikes, mitigation advice and GHG reports for Rwanda's NDC and green finance. It needs field data first. N_2O needs an electrochemical sensor, not yet fitted.

From sensor reading to farmer action

1 · Data inputs

Collar: temperature, pulse, SpO2, motion, GPS. Station: RFID, milk level, TDS. Barn: climate, gases, feed, water.

2 · Edge and cloud

ESP32 and NodeMCU devices post readings every 10 s; cooling and pump rules also run on the device.

3 · Analysis

Today: vet-range rules plus a risk score against each cow's 14-day baseline. Next: trained models.

4 · Farmer action

Alert to farmer, vet or technician: cool the barn, inseminate, treat the cow or hold the milk.

Running today

- Risk score 0–100 per cow: vital-sign ranges plus deviation from her own 14-day average
- Readings tied to each cow by RFID
- Ultrasonic milk volume in litres
- Automatic pump, fan and spray

Model (in development)	Method	Training data
Heat (estrus) detection	Motion and temperature against each cow's own baseline	Collar logs labelled by insemination outcome
Milk-quality flag	TDS and temperature anomalies per cow	Station logs labelled by collection-centre tests
Health risk	Fever, activity drop, lower feed and water intake	Barn and collar logs labelled by vet diagnoses
Emission early warning	Gas, heat and humidity trends	Barn gas logs from pilot farms

The risk score is statistical. No machine-learning model is trained yet: the labelled dataset comes from the field pilot.

Working in the cowshed; accuracy not yet measured

Signal	Rule or method	Field reading (demo video)
Barn climate	Temperature, humidity; fan and spray	28.7–29.2 °C, 51–58%; spray switched on
Barn gases	MQ-5, MQ-135: 60% warn, 80% critical	4–7% of range; uncalibrated
Feed and water	Load-cell feed weight; tank level	≈37 kg feed; tank 52–56%
Milk volume	Ultrasonic level in a 4 L container	0.35–3.6 L read live
Collar vitals	Pulse 48–100 bpm; SpO2 below 90%	60–72 bpm, SpO2 95–96%
Cow risk score	Vital ranges plus 14-day baseline	Flagged a cow live (score 55)
Heat detection	Motion and temperature model	Not yet trained

Versus today: farmers notice fever or heat stress by eye, hours to days later. FarmCare updates every 10 seconds and acts by itself. Accuracy against vet diagnosis and a measuring jug: not yet measured.

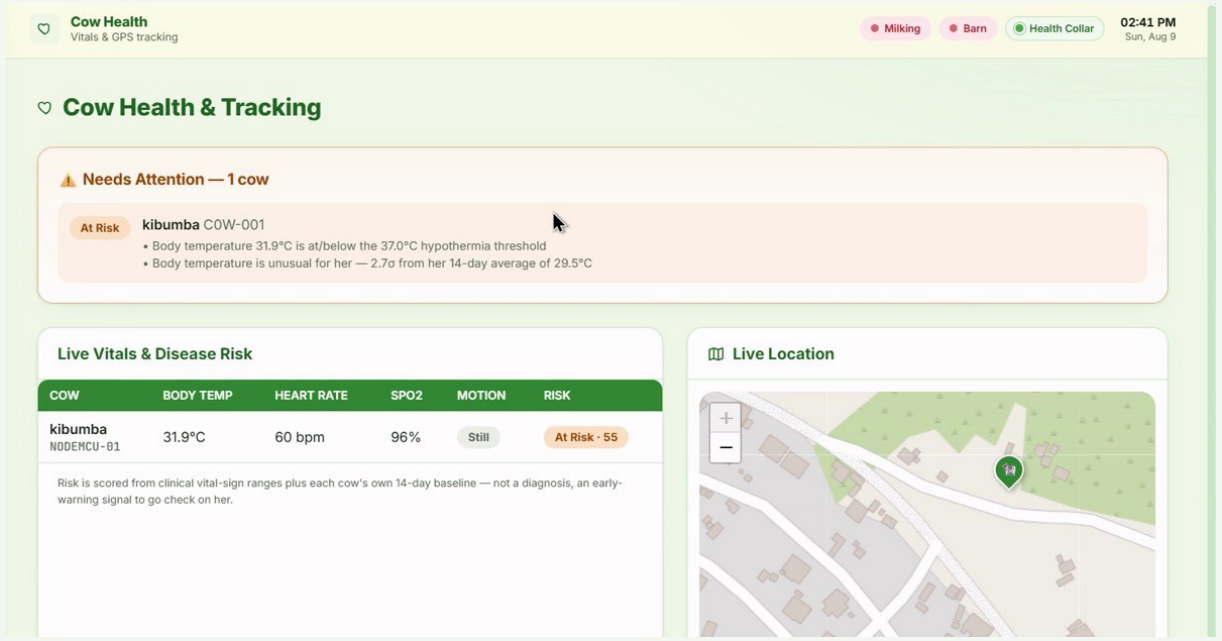
Known limitations

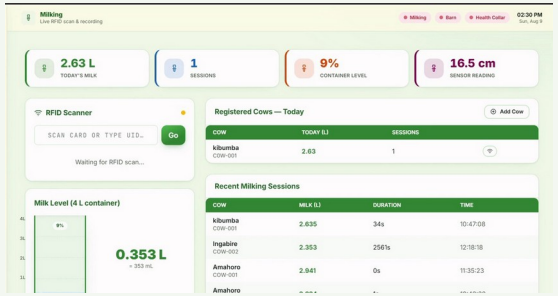
- Collar reads ~32 °C, below core heat: needs better placement
- Gas read as % of range, not ppm
- Pulse sensor designed for humans
- SMS gateway and USSD not yet built
- Kinyarwanda interface not yet built

TRL 6 Working application

Evidence: the full platform is live online and the devices stream from a working cowshed. TRL 7, proof of application, needs a multi-farm pilot over one season.

Two journeys we will show live





Top: Cow Health page. Left: Milking page.

Recorded live in a cowshed in Kanyinya, Nyarugenge.

▶️ **Watch the demo video**

Journey A · The collar flags a cow at risk

1. Collar streams temperature, pulse, SpO2, motion and GPS
2. Risk score compares her with her own 14-day average
3. Kibumba appears under Needs Attention, with reasons
4. Farmer or vet checks her, using her live location

Journey B · Recording milk at the container

1. Farmer scans the cow’s RFID tag
2. Milk goes into the smart container
3. Ultrasonic sensor shows the litres live
4. Record Milk saves the session to her history

Live on the day at farmcareservices.com/app with the prototype streaming; the demo video is our backup.

One pilot farm so far; impact not yet measured

REGISTERED USERS

4

one per role: admin, farmer, vet, technician

PILOT PARTICIPANTS

1

farm: the cowshed in Kanyinya, Nyarugenge

PAYING CUSTOMERS

0

FarmCare is pre-revenue

INTENDED BENEFICIARIES

1–10

cows per farm, five dairy districts

Outcome	Baseline, then method	Sample, period
Conception rate	Paper records, then collar heat alerts	Pilot herd, one season
Milk rejection	Collection-centre records, before and after	Each farm, monthly
Mastitis cases	Vet records, then platform alerts	Every case in the pilot
Milk per cow	Farmer records, then station logs	Each cow, 12 months
Heat stress, barn gases	Climate, gas and cooling logs	Each barn, all year

Design: baseline and endline on each pilot farm, compared with matched farms without kits.

ACHIEVED RESULTS

None yet. Repeat use and farm outcomes are not yet measured.

PROJECTED: HYPOTHESES TO TEST

- More pregnancies per insemination
- Less milk rejected at collection
- Mastitis caught earlier
- More milk per cow, lower emissions per litre

Three revenue streams; pre-revenue today

Hardware kits

Who pays: Farmers, cooperatives; instalments

One-time purchase; ESP32 parts, assembled in Rwanda.

Station ≈ RWF 100k · Barn ≈ RWF 250k
Collar ≈ RWF 85k per cow

Milk-linked subscription

Who pays: Farmers, via their cooperative

Dashboard, alerts, records and maintenance, paid only from milk sold.

3% of milk sold; nothing in dry months
Cap: RWF 10,000 per farm a month

Institutional contracts

Who pays: Inyange, Mukamira, NGOs, banks

Bundled deployments with dairy programmes, plus quality data.

Quoted per deployment

FARMCARE REVENUE
RWF 0
pre-revenue

COMPANY TURNOVER
RWF 3M
last year, informal side projects: AI, IoT design, tech support

GRANTS & INVESTMENT
RWF 0
founder-funded to date

LETTERS OF INTENT
0 signed
first cooperative is next

Primary
Smallholders with 1–10 cows; 1.45–1.73M cattle nationally

Secondary
Cooperatives, collection centres and processors

Third
MINAGRI, RAB, FAO, banks and insurers

Prices are estimates from Kigali part costs plus assembly; to be tested in the pilot. Projected revenue: not yet measured.

Six people across technology, field operations and climate

BJ

Bikorimana Jean Claude

IT SPECIALIST & COORDINATION

Field operations: integration and partnerships

NB

Nindenkimana Blaise

AI ENGINEER

Technical development: platform, AI models, architecture

HE

Harerimana Eric

IOT ENGINEER

Technical development: sensors, RFID and connectivity

ME

Mucyo Enock

PROJECT DIRECTOR

Technical development: machine learning, data security

IE

Izere Egide

ENVIRONMENTAL EXPERT

Agricultural validation: climate-smart fieldwork

NE

Niyonsaba Elina

OPERATIONS & SALES

Field operations: farmer and cooperative onboarding

Gap: veterinary validation

No veterinarian on the team yet. We will secure a veterinary adviser through RAB or the University of Rwanda before the pilot.

Active partners: none signed yet

Targets: dairy cooperatives in five districts, Inyange and Mukamira, RAB and MINAGRI, University of Rwanda and ILRI, FAO and Smart Africa.

Rwanda first, then Smart Africa member states

Phase 1 · Build

Small production run of field-ready kits: boards, sensors and enclosures

Phase 2 · Pilot

First cooperative agreement; field test with a real herd; first trained models

Phase 3 · Scale

More Rwandan dairy zones, then other Smart Africa member states

Local data	Recalibrate thresholds and retrain models on local breeds, climate and farm data
Languages	English today; Kinyarwanda next, then a language pack per country for the dashboard, SMS and USSD
Connectivity	Wi-Fi today; SMS and USSD for basic phones through a local telecom partner
Regulation	Check data-protection law and IoT import rules before the first kit ships
Extension systems	Share herd, milk-quality and emission data with ministry and extension systems through our API

Status: not yet deployed beyond our pilot farm in Kanyinya, Nyarugenge. SMS, USSD and Kinyarwanda are planned, not built.

FarmCare

AI and IoT for climate-smart dairy farming

It already runs

Three devices and a live platform, tested in a working cowshed

Climate-smart

Cools barns in heat, tracks manure gases, more milk per cow

One connected view

Breeding, health, feed and milk quality, linked to the buyer

Ready for the field

Your support takes FarmCare from prototype to a real herd

Thank You

Ready for your questions

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